

# Insider Threat & Data Exfiltration Investigation

Analysis of suspected KYC data exfiltration and anti-forensic activity on a Windows 10 Enterprise LTSC system

CASE REFERENCE

ITI-2026-001

EXAMINER

Paulo Vaz

REPORT DATE

July 12, 2026

REPORT VERSION

1.1 — Final

CLASSIFICATION

Academic Portfolio

TARGET SYSTEM

DESKTOP-2F965T3 (NEXCHAIN-WS01)

EVIDENCE IMAGE (MD5)

7e5404e298b1d886534a2845e4d880bc

STANDARDS REFERENCED

ISO/IEC 27037, 27035, 27041, 27042, 27043 ·  
NIST SP 800-86, 800-61 · RFC 3227 · LGPD

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## 1. Identification

|                                    |                                                                                                                    |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Case Reference                     | ITI-2026-001                                                                                                       |
| Examiner                           | Paulo Vaz                                                                                                          |
| Requesting Party                   | Academic Portfolio — Self-initiated                                                                                |
| Attack Simulation Date             | July 4, 2026                                                                                                       |
| Examination Date (Acquisition)     | July 6, 2026                                                                                                       |
| Examination Date (Memory Analysis) | July 7, 2026                                                                                                       |
| Examination Date (Disk Analysis)   | July 8–10, 2026                                                                                                    |
| Report Date                        | July 12, 2026                                                                                                      |
| Report Version                     | 1.1 — Final                                                                                                        |
| Classification                     | Academic Portfolio                                                                                                 |
| Standards Referenced               | ISO/IEC 27037, 27035, 27041, 27042, 27043; NIST SP 800-86, 800-61; RFC 3227; LGPD (Lei Geral de Proteção de Dados) |

### NOTE ON DATES

The attack simulation was performed on **July 4, 2026**; the target VM was then suspended to a saved state (not powered off) and forensic acquisition was performed on **July 6, 2026** after the VM was resumed. Because the guest was suspended rather than shut down, the memory image's internal clock reflects the July 4 suspension moment — this is expected and is explained in full in Section 6.2.

## 2. Executive Summary

This report documents a digital forensic investigation conducted on a Windows 10 Enterprise LTSC virtual machine (host name **DESKTOP-2F965T3**, designated **NEXCHAIN-WS01** within the fictitious NexChain Exchange environment). The investigation was carried out in a controlled laboratory as part of an academic portfolio project.

The examination sought to determine whether a compliance analyst had exfiltrated customer KYC (Know Your Customer) data — names, national identifiers, and cryptocurrency wallet addresses — to an external device prior to resignation, and whether anti-forensic techniques were used to conceal the activity. The investigation was conducted as a **blind analysis**: the attacker's actions were recorded and sealed in a GPG-encrypted log before the investigation began, and that log was opened only after all findings were finalized, to allow an objective measure of investigative completeness (Section 11).

The investigation spanned eight phases, from attack simulation and forensic acquisition through memory, disk, and anti-forensic analysis to timeline reconstruction and this report. Multiple independent tools were employed — Volatility

3, The Sleuth Kit 4.14.0, and native forensic utilities — with cross-validation between evidence sources used as a methodological control.

### Principal findings

| Finding                                | Detail                                                                                                                                                                                             |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data exfiltration confirmed            | A KYC dataset ( <code>kyc_customers.csv</code> ) was copied from the user's Documents folder to a mounted virtual disk (drive E:) acting as a simulated removable device.                          |
| Three anti-forensic techniques used    | Secure deletion (SDelete), Security event-log clearing (wevtutil), and NTFS timestamp manipulation (timestomping) were all executed and all independently detected.                                |
| Full command history recovered         | The complete PowerShell command history of both user accounts ( <code>analyst01</code> and <code>admin</code> ) was recovered from disk, providing a verbatim, first-party record of every action. |
| Simulated device identified as virtual | Registry analysis established the "USB device" was a Microsoft Virtual Disk attached internally via the OS, not a physical USB device — proven through the SCSI enumeration vendor string.         |
| Log clearing self-recorded             | The act of clearing the Security log generated Event ID 1102, which itself records the responsible account ( <code>admin</code> ) and precise time — the attacker could not suppress this record.  |
| Timestomping structurally proven       | A discrepancy between the NTFS \$STANDARD_INFORMATION and \$FILE_NAME timestamps conclusively demonstrated timestamp falsification, independent of any log.                                        |
| 100% investigative coverage            | Every action recorded in the sealed attacker log was independently recovered by the investigation, most through multiple convergent sources (Section 11).                                          |

### 3. Objectives

The investigation was conducted with the following objectives:

1. Determine whether customer KYC data was copied from the target system to an external device.
2. Identify and recover all disk and memory artifacts related to the suspected exfiltration and any anti-forensic activity.
3. Determine the nature of the external device used (physical vs. virtual).
4. Detect and technically prove any use of anti-forensic techniques (secure deletion, log tampering, timestamp manipulation).
5. Reconstruct a unified, chronologically consistent timeline correlating memory, disk, and event-log evidence.
6. Assess investigative completeness through blind comparison against a pre-sealed record of the attacker's actions.
7. Produce a formal forensic report documenting all findings, methodology, limitations, and procedural deviations.

## 4. Methodology & Standards

The investigation followed a structured eight-phase methodology aligned with international standards for digital evidence handling and incident investigation. Each standard governs a specific dimension of the work:

| Standard       | Application in this investigation                                                                                                                                     |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ISO/IEC 27037  | Identification, collection, acquisition and preservation of digital evidence — governs the forensic imaging and hashing procedures in Phase 02.                       |
| ISO/IEC 27035  | Information security incident management — frames the insider-threat scenario as a security incident with a structured response.                                      |
| ISO/IEC 27041  | Assurance for the suitability and adequacy of the investigative method — supports the multi-tool cross-validation approach.                                           |
| ISO/IEC 27042  | Analysis and interpretation of digital evidence — governs the memory, disk, anti-forensic, and timeline-correlation analysis.                                         |
| ISO/IEC 27043  | Incident investigation principles and processes — governs the end-to-end investigative structure.                                                                     |
| NIST SP 800-86 | Guide to integrating forensic techniques into incident response — informs the acquisition-to-analysis workflow.                                                       |
| NIST SP 800-61 | Computer security incident handling guide — informs the incident-response framing.                                                                                    |
| RFC 3227       | Guidelines for evidence collection and archiving — the order of volatility it defines justifies capturing volatile memory before shutdown, prior to disk acquisition. |
| LGPD           | Brazil's General Data Protection Law — establishes the regulatory significance of the exfiltrated KYC data as personal data (see Sections 5 and 11).                  |

## Phase structure

| Phase                                     | Summary                                                                                                                                                                      |
|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase 00 — Environment Setup              | Two-VM laboratory: Windows 10 Enterprise LTSC target and Kali Linux 2026.1 investigator, connected via host-only network; clean baseline snapshot taken before any activity. |
| Phase 01 — Attack Simulation              | Controlled generation of the incident from the attacker's perspective; actions sealed in a GPG-encrypted log for later blind comparison.                                     |
| Phase 02 — Forensic Acquisition           | Memory dump captured pre-shutdown (order of volatility, RFC 3227); disk acquired to E01 via ewf_acquire with MD5 + SHA-256 verification.                                     |
| Phase 03 — Memory Analysis                | Volatility 3 analysis of the 8.7 GB memory image; recovery of process list and command-history fragments.                                                                    |
| Phase 04 — Disk Analysis                  | The Sleuth Kit analysis of the E01: MFT, USN Journal, Prefetch, Event Log, Registry, and NTFS timestamp analysis.                                                            |
| Phase 05 — Anti-Forensic Analysis         | Evaluation of each anti-forensic technique's effectiveness against what the investigation recovered.                                                                         |
| Phase 06 — Timeline Reconstruction        | Correlation of all timestamped events across sources into one UTC-normalized timeline.                                                                                       |
| Phase 07 — Report & Sealed Log Comparison | Blind-analysis comparison and this report.                                                                                                                                   |

## 5. Case Description

The target system is a Windows 10 Enterprise LTSC 21H2 virtual machine (build 10.0.19044) with 8 GB RAM, 4 virtual CPUs, and a 50 GiB NTFS system partition. Two local user accounts are relevant to the investigation:

|                        |                                                                                                                            |
|------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <code>analyst01</code> | Standard user — represents the suspected insider (a compliance analyst). SID S-1-5-21-399538211-3085301091-319685644-1003. |
| <code>admin</code>     | Local administrator — used for actions requiring elevation (virtual disk creation, event-log clearing).                    |

The scenario models a compliance analyst at the fictitious NexChain Exchange suspected of exfiltrating customer KYC data prior to resignation. The KYC dataset contained names, Brazilian national identifiers (CPF), and cryptocurrency wallet addresses — a category of personal and financial data whose unauthorized disclosure falls within the scope of the LGPD. The analyst allegedly copied the data to an external device and attempted to conceal the activity using anti-forensic techniques.

The investigation was initiated as an academic exercise demonstrating end-to-end DFIR methodology — from evidence generation and forensic acquisition through memory and disk analysis to formal reporting — using real forensic tools and industry-standard procedures.

## 6. Evidence & Chain of Custody

### 6.1 Primary Disk Image

|                         |                                                                  |
|-------------------------|------------------------------------------------------------------|
| Description             | Windows 10 Enterprise LTSC 21H2 system disk (post-attack state)  |
| Source                  | Consolidated clone of the "post-attack" VirtualBox snapshot      |
| Image format            | Expert Witness Format (E01), 35 segments                         |
| Logical size            | 50 GiB (53,687,091,200 bytes)                                    |
| Acquisition tool        | ewf_acquire 20140816                                             |
| Acquisition date        | July 6, 2026                                                     |
| MD5 (stored / verified) | 7e5404e298b1d886534a2845e4d880bc (match)                         |
| SHA-256                 | 3d61d7774ca4f85a88077d46ce535debe463f341aff2404cc885dc3471886188 |
| Partition offset (TSK)  | 104448 (primary NTFS)                                            |
| Status                  | Active — authoritative image for all disk analysis               |

### 6.2 Memory Image

|                  |                                                                  |
|------------------|------------------------------------------------------------------|
| Description      | Full RAM capture of the target VM                                |
| File             | memdump.elf (ELF core dump)                                      |
| Acquisition tool | VBoxManage debugvm ... dumpvmmcore                               |
| Size             | 8,730,336,996 bytes (~8.7 GiB)                                   |
| Acquisition date | July 6, 2026 (after VM resume from saved state)                  |
| SHA-256          | 9e2a212c3a63e484cde27151467fcaa75b9a13aaf3945707b8c8bca2ce14a90e |
| Status           | Secondary evidence — corroborates disk findings                  |

#### NOTE ON THE MEMORY IMAGE ACQUISITION DATE

The attack simulation was performed on **July 4, 2026**. Rather than shutting the target VM down, it was **suspended to a saved state**, preserving volatile memory intact, and resumed on **July 6, 2026**, when the memory dump was captured (before shutdown) and the disk was imaged. Consequently the memory image's internal **SystemTime** reads **2026-07-04 16:10:21 UTC** — the guest clock reflects the moment of suspension on the attack-simulation day, which is why the image's internal timestamp legitimately predates its July 6 acquisition date. This also explains why the interactive PowerShell command history from the July 4 session survived in RAM and was recoverable in Phase 03: suspension preserves memory, whereas a shutdown would have cleared it. The full sequence is recorded in the chain-of-custody document.

### 6.3 Sealed Attacker Log

The attacker's actions were documented in real time during Phase 01 and immediately sealed with GPG (AES-256) symmetric encryption before the investigation began. The encrypted file remained sealed until all investigative findings were finalized, at which point it was decrypted for the blind-analysis comparison in Section 11. This procedure preserves investigator objectivity and provides an auditable measure of investigative completeness.

**PROCEDURAL NOTE — ACQUISITION DESTINATION**

Two initial disk-acquisition attempts failed mid-write due to I/O instability in the VirtualBox shared-folder (vboxsf) layer used as the initial output target. The acquisition was successfully completed to a dedicated, locally-attached virtual disk instead. No evidence was lost or altered; the failed partial files were discarded before the successful acquisition. Full detail is recorded in the chain-of-custody document. This deviation would warrant a certified hardware write-blocker and dedicated evidence storage in a production context.

## 7. Forensic Timeline

The following unified timeline was reconstructed from independent evidence sources spanning memory, disk metadata, and the Windows event log. All timestamps are normalized to UTC, the standard reference frame for forensic reporting, so that events originally recorded in different time zones (host log in BRT/UTC-3; system artifacts in UTC; TSK output in EDT/UTC-4) sit on one comparable axis.

| UTC      | Original | Zone | Event                                             | Source          | Type         |
|----------|----------|------|---------------------------------------------------|-----------------|--------------|
| ~14:45   | 11:45    | BRT  | VHD created and mounted as E:                     | Sealed log      | attacker     |
| 14:50:45 | 10:50:45 | EDT  | kyc_customers.csv created in Documents            | \$FN via istat  | system       |
| ~15:06   | 12:06    | BRT  | File copied to E:\ (exfiltration)                 | Sealed log      | attacker     |
| ~15:16   | 12:16    | BRT  | SDelete run on original file                      | Sealed log      | attacker     |
| 15:20:50 | 15:20:50 | UTC  | PowerShell session #1 started                     | pslist (memory) | system       |
| 15:20:54 | 15:20:54 | UTC  | Security log cleared — Event ID 1102              | Event Log       | system       |
| 15:26:06 | 15:26:06 | UTC  | PowerShell session #2 started                     | pslist (memory) | system       |
| 15:26:30 | 11:26:30 | EDT  | Timestomping applied (MFT Modified)               | istat \$SI      | attacker     |
| 16:10:21 | 16:10:21 | UTC  | Memory dump captured (guest clock, at suspension) | windows.info    | investigator |

**NOTE ON TIME-ZONE NORMALIZATION**

Artifacts from the same Windows system surface here in different zones (Event Log in UTC, TSK *istat* in EDT) — this reflects how different subsystems store and how different tools render time, not a clock inconsistency. The zone conversions were validated using a cross-source anchor: the *wevtutil* action appears in the hand-written log at 12:21 (BRT) and in Event ID 1102 at 15:20:54 (UTC), a ~3-hour offset confirming BRT = UTC-3. Times marked "~" derive from the manually-written attacker log and are approximate to the minute; where they diverge from system-generated timestamps, the system source is authoritative. The 16:10:21 UTC memory-capture row is the guest's internal clock at the moment of suspension (July 4); the wall-clock acquisition occurred July 6 (Section 6.2).



Reading the timeline as a narrative, the operation reconstructs cleanly: preparation (VHD creation) → data staging (KYC file creation) → exfiltration (copy to E:) → a concentrated anti-forensic phase (SDelete, log clearing, and timestomping within a ~10-minute window, 15:16–15:26 UTC) → and finally the investigator's memory capture. The concentration of all three anti-forensic techniques into a single short window is itself an analytical indicator of a deliberate, staged clean-up rather than scattered, reactive concealment.

## 8. Technical Analysis

### 8.1 Memory Analysis (Phase 03)

The 8.7 GB memory image was analyzed with Volatility 3 Framework 2.28.1. The system profile was confirmed as Windows 10 build 19041, consistent with the disk image.

**Process list (windows.pslist).** Two distinct PowerShell sessions were active at capture time: one started at 15:20:50 UTC (spawned by a RuntimeBroker process) and one at 15:26:06 UTC (spawned by explorer.exe — the expected parent for a manually opened terminal). Neither `sdelete64.exe` nor `wevtutil.exe` appeared in the process list, consistent with both having already exited by capture time — expected for short-lived command-line tools.

**Command history in memory.** A raw string search of the memory image recovered fragments of the attacker's interactive PowerShell history — including the literal SDelete command line and the file paths involved — from the PSReadLine console buffer retained in RAM. This established, from volatile memory alone, that the deletion and copy operations had occurred, even though the responsible processes were no longer running and the original file had been removed from disk.

**File scan.** A single file-object reference to `kyc_customers.csv` remained resident in memory, consistent with the copy present on the simulated USB device (E:) at capture time.

**Network state (windows.netstat).** Three established TCP connections were present, all on the NAT interface and directed to Microsoft Azure address ranges (routine telemetry/update traffic). No evidence of network-based exfiltration was found — consistent with the physical-media exfiltration method established elsewhere.

### 8.2 Disk Analysis (Phase 04)

The E01 image was mounted read-only at every layer and analyzed with The Sleuth Kit 4.14.0.

**8.2.1 Deleted-file recovery and the USN Journal.** The original `kyc_customers.csv` in the user's Documents folder was not recoverable via the Master File Table (MFT): three progressively broader `fls` searches returned no entry — not even a deleted or orphaned record. This is a stronger anti-forensic result than a standard deletion, which typically leaves a recoverable MFT entry; SDelete's secure-overwrite behavior invalidated the entry itself. However, the file's prior existence was proven through the USN Journal ( `$Extend\UsnJrnl:$J` ), an NTFS change-log mechanism architecturally separate from the MFT. A UTF-16LE string search of the extracted journal returned ten occurrences of `kyc_customers.csv`, confirming the file existed and underwent file-system activity despite being unrecoverable from the MFT.

**8.2.2 Prefetch — tool execution.** Windows Prefetch caches an entry for every executed program, independent of whether the process still exists. Two entries confirmed execution of both anti-forensic tools:

| Prefetch file             | Last execution (local, EDT)    |
|---------------------------|--------------------------------|
| SDELETE64.EXE-6548E1FA.pf | 2026-07-04 11:11 (→ 15:11 UTC) |
| WEVTUTIL.EXE-EF5861C4.pf  | 2026-07-04 11:20 (→ 15:20 UTC) |

These Prefetch modification times are rendered by the investigator VM in its local zone (EDT/UTC-4); converted to UTC they corroborate independently-dated events (Event ID 1102 and the sealed log) to within one to five minutes. They are used only to confirm that each tool executed, not as a primary timing source.

**8.2.3 Windows Event Log — the self-recording log clear.** The `Security.evtx` log was extracted and parsed. Its small size and the fact that its first surviving event is the clearing action itself both indicate the log was reset. That clearing action is recorded by Event ID 1102 — an event Windows generates automatically whenever the audit log is cleared, which cannot be suppressed by the clearing command itself:

|                     |                                    |
|---------------------|------------------------------------|
| Event ID            | 1102 — "The audit log was cleared" |
| Timestamp           | 2026-07-04 15:20:54.560593 UTC     |
| Responsible account | admin                              |
| Client Process ID   | 7812                               |

Process-creation auditing (Event ID 4688) was not enabled in the target's default configuration, so SDelete's process creation was not captured in the Security log. This is an expected consequence of the default Windows audit policy and is noted as an environmental limitation rather than an investigative gap; execution was independently confirmed via Prefetch and command history.

**8.3 Registry Analysis — nature of the external device.** The nature of the "USB device" was established through a three-key elimination process in the SYSTEM hive:

- **USBSTOR** — this key did not exist. It is populated only when a device enumerates over the USB bus, so its absence rules out a physical or passed-through USB device.
- **MountedDevices** — recorded drive letter E: mapped to a disk signature distinct from the CD-ROM/optical entries, confirming E: was a real, separately-attached disk.
- **SCSI enumeration** — contained the entry `Disk&Ven_Msft&Prod_Virtual_Disk`. The vendor string "Msft" (Microsoft) identifies this as a virtual disk created by the Windows Virtual Disk Service (the mechanism behind `diskpart create vdisk`), in contrast to the VM's real disk which enumerates as `Ven_VBOX`.

The combination — absent from USB, present in SCSI under a Microsoft vendor string — conclusively identifies the exfiltration medium as a virtual disk (VHD) attached internally via the operating system, not a physical USB device.

**PowerShell command history — the primary artifact.** PowerShell's PSReadLine module persists console input history to disk, per user profile. Both relevant profiles retained complete histories: `analyst01` held the full attack session (SDelete download and setup, KYC file creation including a first failed attempt preserved verbatim, the copy to E:, the SDelete execution, and the complete three-line timestamping sequence); `admin` held two lines (`diskpart virtual-disk creation` and `wevtutil cl Security`). This is the single strongest artifact in the investigation — a persistent, unmediated, first-party record. Because each account maintains its own history, actions could be attributed to a specific account, establishing not merely what occurred but who performed each step.

**NTFS timestamp analysis — timestamping.** NTFS maintains two independent timestamp sets per file: `$STANDARD_INFORMATION` (\$SI), modifiable via standard Windows APIs and displayed by Explorer and PowerShell; and `$FILE_NAME` (\$FN), stored in the parent directory index and not exposed to the same modification path. Examination of the exfiltrated file (via `istat`) showed exactly the expected discrepancy:

| Attribute set          | Created             | Interpretation                          |
|------------------------|---------------------|-----------------------------------------|
| \$STANDARD_INFORMATION | 2024-01-01 06:00:00 | Falsified value planted by the attacker |
| \$FILE_NAME            | 2026-07-04 10:50:45 | Real creation time, untouched           |

The discrepancy is conclusive, structural proof of timestamp falsification — and, critically, it does not depend on any log, memory image, or command history surviving. Additionally, the \$SI "MFT Modified" sub-field (2026-07-04 11:26:30) retained the real time of the timestamping action, because that field is updated automatically by the file system and was not settable via the technique used — providing a further corroboration point.

## 9. Anti-Forensic Analysis

Three anti-forensic techniques were used. In every case the technique achieved its narrowest data-level objective but failed to conceal that the activity had occurred, because each was independently detected through multiple convergent sources.

| Technique               | Attacker's goal                       | What succeeded                               | What exposed it                                                          |
|-------------------------|---------------------------------------|----------------------------------------------|--------------------------------------------------------------------------|
| SDelete                 | Prevent recovery of the original file | MFT-level recovery blocked (content + entry) | USN Journal, Prefetch, PowerShell history, memory strings                |
| wevtutil cl<br>Security | Erase the audit trail                 | Prior log content destroyed                  | Event ID 1102 (self-recording), Prefetch, PowerShell history             |
| Timestamping            | Disguise the file's true date         | \$SI (user-facing) successfully falsified    | \$FN retained real time; \$SI MFT-Modified sub-field; PowerShell history |

**Overall assessment.** All three techniques achieved their literal objective (destroy this specific data point) but none achieved the broader goal of concealing that anti-forensic activity took place. In each case at least three independent artifact categories — spanning memory, disk metadata, and persistent command history — converged on the same conclusion. Notably, the log-clearing technique is arguably counter-productive for the attacker: it destroys prior content but generates Event ID 1102, which draws direct attention to the exact moment concealment began.

## 10. Evidence Reliability Assessment

Not all evidence carries the same evidentiary weight. Following standard forensic classification, each item is graded as **PRIMARY** (a direct, first-party record of the act itself), **CORROBORATIVE** (independently supports a conclusion but is not, alone, the act being proven), or **CIRCUMSTANTIAL** (consistent with a conclusion but does not, by itself, establish it). This classification — recommended practice under ISO/IEC 27041 — makes explicit which findings could stand alone under challenge and which depend on convergence with other sources.

| Evidence                                                     | Classification | Basis for classification                                                                                                                                       |
|--------------------------------------------------------------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PowerShell command history<br>(ConsoleHost_history.txt)      | PRIMARY        | A verbatim, unmediated record written by the shell itself at the moment each command was typed — no reconstruction or interpretation by the examiner.          |
| Event ID 1102 (Security log clear)                           | PRIMARY        | Automatically generated by the Windows audit subsystem, not by the examiner or any tool under attacker control — a first-party system record of the act.       |
| NTFS \$SI vs \$FN discrepancy<br>(istat)                     | PRIMARY        | A direct structural read of file-system metadata; the discrepancy itself is the fact being established, not an inference from a related artifact.              |
| USN Journal entries                                          | CORROBORATIVE  | Confirms the file existed and underwent activity, but does not by itself identify which command caused each entry.                                             |
| Prefetch (.pf files)                                         | CORROBORATIVE  | Confirms an executable ran and when it last ran, but not what arguments were used or who initiated it.                                                         |
| Memory strings (Volatility)                                  | CORROBORATIVE  | Recovered via broad string search rather than structured parsing of a known data structure; consistent with the command history but reconstructed, not native. |
| Registry SCSI vendor string<br>(Ven_Msft)                    | CORROBORATIVE  | Strongly indicative of a virtual disk, but relies on interpreting a vendor-identification convention rather than a direct assertion of device type.            |
| PowerShell session parent process<br>(RuntimeBroker anomaly) | CIRCUMSTANTIAL | Atypical but not independently conclusive of anything; noted for awareness, not relied upon in any finding.                                                    |

The investigation's central conclusions — exfiltration, the three anti-forensic techniques, and their attribution to specific accounts — are each supported by at least one primary source, with multiple corroborative sources in every case. No conclusion in this report rests on circumstantial evidence alone.

## 11. Blind-Analysis Comparison (Sealed Log)

After all findings were finalized, the GPG-sealed attacker log from Phase 01 was decrypted and compared against what the investigation had independently recovered. The purpose of this comparison is to provide an objective, auditable measure of investigative completeness — what was found, what was missed, and why.

| # | Attacker action (sealed log) | Independently recovered? | Corroborating sources                                       |
|---|------------------------------|--------------------------|-------------------------------------------------------------|
| 1 | VHD mounted as E:            | Yes                      | Registry (MountedDevices, SCSI), PowerShell history (admin) |
| 2 | File copied to E:\           | Yes                      | USN Journal, PowerShell history (analyst01), memory         |
| 3 | SDelete on original file     | Yes                      | Prefetch, PowerShell history, USN Journal, memory           |
| 4 | wevtutil cl Security         | Yes                      | Event ID 1102, Prefetch, PowerShell history (admin)         |
| 5 | Timestamping applied         | Yes                      | \$SI vs \$FN (istat), PowerShell history (analyst01)        |

**Result: 100% of the recorded attacker actions were independently recovered**, most through multiple convergent sources. No action escaped detection entirely.

## A measured view of the result

Complete recovery in this case should be read with appropriate context, not as evidence that anti-forensics is futile in general:

- **Evidence quality varied by action, even where detection was complete.** Some actions were confirmed by four independent sources; others by two or three. Against a more thorough adversary — one who also cleared the PowerShell history (via Clear-History and deletion of ConsoleHost\_history.txt) — several actions would have rested on weaker evidence, and the single strongest artifact in this investigation would have been absent.
- **The investigation had visibility only from the first recorded action onward.** The sealed log begins at VHD creation; the analysis cannot speak to reconnaissance, decision-making, or preparation that may have preceded it.
- **Technical evidence establishes acts, not motive.** The investigation proves what occurred, when, and by which account, with high confidence. It does not establish intent, whether the actor operated alone or under direction, or what was ultimately done with the exfiltrated data — a structural limit of any technical examination.

## 12. Legal Admissibility Context

While this report is an academic exercise and does not constitute a legal instrument (Section 16), the methodology applied throughout was deliberately chosen to reflect practices that support evidentiary admissibility in a real proceeding. Four factors are particularly relevant:

- **Integrity and non-repudiation of the source.** The disk image was acquired via a forensically sound method (ewf\_acquire, EnCase format) with dual-hash verification (MD5 and SHA-256, both independently re-verified), and all subsequent analysis was performed against read-only mounts. This chain — acquire, hash, verify, analyze without altering the source — is the foundation courts and opposing experts typically scrutinize first.
- **Reproducibility.** Every finding in this report is tied to a specific, documented command against a specific, hashed evidence file. In principle, a second examiner with access to the same E01 and memory image could repeat every step in Sections 8–11 and arrive at the same result — a core requirement for evidence to withstand challenge (aligned with ISO/IEC 27041, assurance of method adequacy).

- **Multi-source corroboration.** As formalized in Section 10, the central conclusions of this case rest on primary evidence corroborated by multiple independent sources, rather than on a single artifact. This reduces reliance on any one tool's correctness and mirrors how expert evidence is typically strengthened for cross-examination.
- **Documented chain of custody and disclosed deviations.** Every procedural deviation from ideal practice (Section 14) — including the shared-folder acquisition failure and the saved-state acquisition timing (Section 6.2) — is disclosed rather than omitted. Undisclosed deviations, if later discovered, are far more damaging to admissibility than disclosed ones; transparency about limitations is itself a credibility-preserving practice.

Under Brazilian procedure, this kind of technical examination would typically inform a *laudo pericial* prepared by a court-appointed or party-retained expert (*perito*), governed by the Código de Processo Penal / Código de Processo Civil rather than by this report's academic format. The forensic discipline demonstrated here — hash-verified acquisition, reproducible steps, and multi-source corroboration — is the same discipline that would underpin such a formal instrument.

## 13. Conclusions

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Based on the totality of evidence across memory and disk analysis, the following conclusions are established with high confidence:

1. **Customer KYC data was exfiltrated to an external device.** The dataset was copied from `C:\Users\analyst01\Documents` to drive E:, confirmed by the PowerShell command history (both in memory and on disk), the USN Journal, and a memory-resident file object.
2. **The external device was a virtual disk attached via the operating system.** Registry analysis (absence in USBSTOR; presence in SCSI as Ven\_Msft Virtual\_Disk) establishes the medium was a VHD attached internally, not a physical USB device.
3. **Three anti-forensic techniques were used and all were detected.** Secure deletion, Security-log clearing, and timestomping were each independently confirmed; none concealed the fact that the activity occurred.
4. **Individual actions are attributable to specific accounts.** Per-user PowerShell histories attribute exfiltration, secure deletion, and timestomping to analyst01, and virtual-disk creation and log clearing to admin.
5. **The timeline is internally consistent across all sources.** Memory, disk metadata, and event-log timestamps correlate coherently once normalized to UTC; the zone conversion was validated against a cross-source anchor.
6. **The incident constitutes a personal-data breach within the scope of the LGPD.** The exfiltrated data (names, CPF, wallet addresses) is personal data; in a real environment this would trigger obligations including risk assessment and potential notification to the national authority (ANPD) and affected data subjects.

## 14. Limitations

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Each limitation below is stated together with its concrete impact on the report's conclusions, so that its practical significance — rather than only its existence — is clear.

- **No hardware write blocker.** The investigation used a virtualized environment without a physical write blocker. Integrity was maintained through read-only mounts and hash verification. *Impact:* in a real proceeding this would likely draw scrutiny; here it is mitigated by the dual-hash verification (Section 6) showing the acquired image matches its stored hash, and by every subsequent analysis command being run against read-only-mounted evidence.

- **Shared-folder acquisition instability.** The initial acquisition destination (VirtualBox shared folder) proved unstable for sustained writes; acquisition was completed to a dedicated local disk. *Impact:* none on the conclusions — the failed partial files were discarded before the successful acquisition began, and the final image's hash was independently verified twice (Section 6). This is a lab-infrastructure limitation, not an evidentiary one.
- **Saved-state acquisition timing.** The target VM was suspended on July 4 and acquired on July 6, so the memory image's internal clock predates its acquisition date (Section 6.2). *Impact:* none on the conclusions — suspension preserves memory intact (which is why the command history survived), and the divergence between the guest clock and the acquisition date is fully documented rather than a source of ambiguity.
- **Process-creation auditing disabled.** Event ID 4688 was not enabled in the target's default configuration. *Impact:* removes one potential corroboration source for SDelete/wevtutil execution timing, but does not weaken any conclusion — both tools' execution is independently confirmed via Prefetch and command history, so no finding depends on the missing 4688 events.
- **Memory dump hashed with SHA-256 only.** No MD5 was generated for the memory image at acquisition time. *Impact:* minimal — SHA-256 alone is cryptographically sufficient for integrity verification; the omission is a process-consistency gap (the disk image has both hashes) rather than a reliability gap in the memory evidence itself.
- **Controlled lab environment.** Evidence was generated by the examiner in a single-investigator lab. *Impact:* the findings demonstrate methodological competence rather than analysis of independently sourced, adversarial evidence; conclusions about this incident are unaffected, but the exercise cannot speak to how the same methodology would perform against evidence the examiner did not generate.
- **Scope of technical evidence.** As noted in Section 11, the analysis establishes acts and attribution but not motive, intent, or downstream use of the exfiltrated data. *Impact:* directly bounds Conclusions 1–5 to what the evidence can support — acts, timing, and attribution — and explains why no conclusion makes any claim about attacker motive or intent.

## 15. Recommendations

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The following recommendations address both the investigative process and the defensive posture that would improve detection and prevention of comparable incidents:

1. **Enable process-creation auditing (Event ID 4688) with command-line logging.** This would have captured SDelete and wevtutil execution directly in the Security log, providing an additional independent source.
2. **Forward logs to a remote SIEM in real time.** Local log clearing (wevtutil) is only effective because logs reside solely on the host. Real-time forwarding renders local clearing ineffective and would preserve the pre-clear Security events destroyed in this incident.
3. **Restrict local administrator rights and virtual-disk creation.** Both the log clearing and the VHD creation required elevation; constraining administrative rights on analyst workstations raises the barrier to these actions.
4. **Deploy DLP and removable-media controls.** Data-loss-prevention monitoring of KYC data locations, and policy controls on mounting removable or virtual disks, would detect or prevent the exfiltration channel used here.
5. **Use hardware write blockers and dedicated evidence storage.** In any production investigation, acquire through a certified write blocker and write evidence to dedicated, verified storage rather than shared folders.
6. **Generate dual hashes (MD5 + SHA-256) for every acquired artifact at acquisition time, including memory images.**



## 16. Examiner Declaration

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I, Paulo Vaz, hereby declare that:

1. This report accurately reflects the findings of the digital forensic examination conducted on the evidence described herein.
2. The examination was performed using recognized forensic tools and methodologies aligned with ISO/IEC 27037, 27035, 27041, 27042, 27043; NIST SP 800-86 and 800-61; and RFC 3227.
3. All procedural deviations from standard forensic practice have been documented, justified, and disclosed within this report and the chain of custody.
4. The findings, conclusions, and opinions expressed are my own and reflect an honest assessment of the evidence examined.
5. This report was prepared for academic portfolio purposes and does not constitute a legal instrument in any jurisdiction.

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Paulo Vaz — Digital Forensic Examiner

Date: \_\_\_\_\_ / \_\_\_\_\_ / 2026

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Generated July 12, 2026. Evidence image MD5: 7e5404e298b1d886534a2845e4d880bc. Case reference: ITI-2026-001. The NexChain Exchange organization, user accounts, and all customer data referenced are fictitious; this is an academic portfolio exercise.